A Primer on Rheumatologic Laboratory Tests

What They Mean and When to Order Them

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KEYWORDS

• ANA • ANCA • CCP • CRP • ESR • RF

KEY POINTS

- Laboratory tests in rheumatology should be interpreted depending on the clinical scenario because they are rarely diagnostic of any particular disease.
- A positive antinuclear antibody can be found in normal individuals and in patients with other inflammatory and autoimmune diseases and is not diagnostic of systemic lupus erythematosus.
- Rheumatoid factor (RF) and anticyclic citrullinated peptide (anti-CCP) antibodies support the diagnosis of rheumatoid arthritis, and anti-CCP is more specific than RF.
- A positive antineutrophilic cytoplasmic antibody test with antiproteinase 3 or antimyeloperoxidase specificity supports the clinical diagnoses of systemic necrotizing vasculitis but is not diagnostic. A biopsy is necessary to confirm the diagnosis.

INTRODUCTION

The diagnosis of a rheumatologic disease is challenging and requires careful integration of a patient's symptoms, physical examination findings, and results of diagnostic tests. A thorough history and physical examination are the best screening tests in the initial evaluation of a rheumatologic disease, and the results of laboratory tests should be used only to further refine the diagnosis. Many tests are now available that may aid in rapid diagnosis, but false positive results may lead to inappropriate therapy and unnecessary health care expenses. Laboratories perform different assays, and this can be a source of discrepancy and is often the reason that some laboratory tests are repeated when people are seen by rheumatologists. Hence, results should be interpreted judiciously based on their test characteristics in the appropriate clinical context.[1]

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CONSIDERATIONS WHEN INTERPRETING LABORATORY TESTS

Clinicians should have some familiarity with the terms used for interpretation of a test's characteristics (**Table 1**).[1]

The important issue in deciding whether to use a diagnostic test in a particular patient is whether the posttest probability will be significantly different from the pretest probability, given a positive or negative test result.[1] The pretest probability and the posttest probability are the probabilities of the presence of a particular disease before and after ordering a diagnostic test, respectively. The pretest probability is the rough estimate that a clinician makes based on clinical presentation and the prevalence of the disease in question. **Table 2** lists the overall prevalence of selected rheumatologic diseases in the United States.

The sensitivity and specificity of a test are not sufficient to calculate the probability of disease for a given patient. The likelihood ratio (LR) is meant to provide an additional measure that can be of greater value in daily practice. LRs allow the clinician to calculate the posttest probability based upon the pretest probability and the test result. Tests are recommended as being very useful, useful, or not useful based on their associated LRs (**Table 3**).[1]

For example, a 29-year-old African American woman presents with joint swelling, oral ulcers, alopecia, and a facial rash. The clinician thinks her chances of having systemic lupus erythematosus (SLE) are greater than that of the general population and estimates that the patient's pretest probability of SLE is 10%. The clinician wants to order a test (such as an antinuclear antibody [ANA] or related autoantibody test) to confirm this suspicion. If the test has a high positive LR (eg, 10) and the test result is positive, then the posttest probability will be greatly increased. If the LR of the

Table 1
Calculation for sensitivity, specificity, positive predictive value, negative predictive value, and likelihood ratios

Test Result	Disease Present	Disease Absent
Positive	a	b
Negative	c	d

False positive	Positive test result in a patient without the disease
False negative	Negative test result in a patient with the disease
Sensitivity	$\frac{\text{Number of persons with a positive test with the disease (a)}}{\text{Total number of persons who have the disease (a+c)}}$
Specificity	$\frac{\text{Number of persons with a negative test without the disease (d)}}{\text{Number of persons without the disease (b+d)}}$
Positive Predictive value Probability that subjects with a positive screening test truly have the disease (depends on prevalence)	$\frac{\text{Number of persons with a positive test with the disease (a)}}{\text{Number of persons with a positive test (a+b)}}$
Negative predictive value Probability that subjects with a negative screening test truly do not have the disease (depends on prevalence)	$\frac{\text{Number of persons with a negative test without the disease (d)}}{\text{Number of persons with a negative test (c+d)}}$
Positive LR+ (Sensitivity)/(1 – Specificity)	$\frac{\text{Probability of an individual } \textit{with} \text{ the disease having a positive test}}{\text{Probability of an individual } \textit{without} \text{ the disease having a positive test}}$
Negative LR− (1 – Sensitivity)/Specificity	$\frac{\text{Probability of an individual } \textit{with} \text{ the disease having a negative test}}{\text{Probability of an individual } \textit{without} \text{ the disease having a negative test}}$

Table 2
Prevalence of selected rheumatologic diseases

Disease	Prevalence in the United States (per 100,000)
Giant cell arteritis	278
Polymyalgia rheumatic	739
RA	600
SLE	53.6
Systemic sclerosis	27.6

Data from Helmick CG, Felson DT, Lawrence RC, et al. Estimates of the prevalence of arthritis and other rheumatic conditions in the United States. Part I. Arthritis Rheum 2007;58:15–25; and Lawrence RC, Felson DT, Helmick CG, et al. Estimates of the prevalence of arthritis and other rheumatic conditions in the United States. Part II. Arthritis Rheum 2008;58:26–35.

test were slightly less (eg, 5), there would still be a substantial increase in the posttest probability that could be clinically relevant. However, if the test had a small positive LR (eg, 1.2), then the posttest probability will not differ substantially from the pretest probability, and a positive test result would not help the clinician in making a diagnosis.[1] Hence, obtaining an ANA test with an LR of 2.2 would be a useful test in this patient, because posttest probability would be 20% (**Fig. 1**).

Similarly, in a 71-year-old man presenting with fatigue and joint pain, the pretest probability of SLE would be expected to be only slightly greater than the prevalence of SLE in the general population (approximately 0.1% or less). For this patient, even if the test had a greater positive LR of 5, the posttest probability would not be expected to be significantly different from the pretest probability.[1] Hence obtaining an ANA test would not be useful in this patient because the posttest probability would be only 0.2% (see **Fig. 1**).

Antinuclear Antibodies

The ANA test is one of the most frequently ordered tests to evaluate for autoimmune disease, but it can also be detected in nonrheumatic diseases, infections, malignancies, and healthy persons.[2] More than 32 million persons in the United States have a positive ANA test, and the prevalence is higher in women, older individuals, and African Americans.[3]

Methods

- Indirect Immunofluorescence assays (IIFA)
- Enzyme-linked immunosorbent assays (ELISA)
- Multiplex bead assays

Table 3
Likelihood ratios

Positive LR	Negative LR	Clinical Implications
>10	<0.1	Large impact on posttest probability, very useful test
5–10	0.1–0.2	Modest clinical difference, very useful test
2–5	0.2–0.5	Small but relevant clinical difference useful test
<2	>0.5	Rarely any clinical difference, not useful test
1	1	No difference between pretest and posttest probability

Modified from American College of Rheumatology Ad Hoc Committee on Immunologic Testing Guidelines. Guidelines for immunologic laboratory testing in the rheumatic diseases: an introduction. Arthritis Rheum 2002;47:432; with permission.

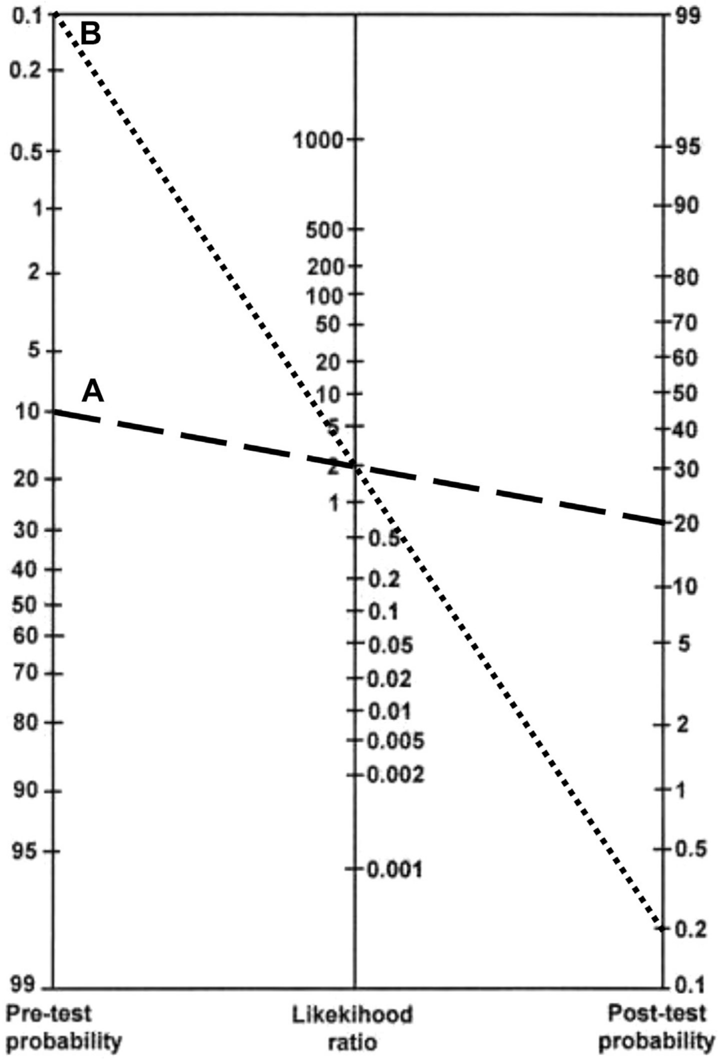

Fig. 1. LR nomogram for patients suspected of having SLE. (*A*) A 29-year-old African American woman with joint swelling, oral ulcers, alopecia, and facial rash. (*B*) A 71-year-old man with fatigue and joint pain. The positive likelihood of SLE in a person with a positive ANA is 2. The 2 examples described in the text are graphed on the nomogram. The estimated pretest probability of case A is 10%. If the woman has a positive ANA, to determine the posttest probability, a line is drawn from the 10% pretest probability through the positive LR of 2. The posttest probability is therefore 20%. For case B with a pretest SLE probability of 0.1%, a positive ANA would translate into a posttest probability of 0.2. (*Modified from* American College of Rheumatology Ad Hoc Committee on Immunologic Testing Guidelines. Guidelines for immunologic laboratory testing in the rheumatic diseases: an introduction. Arthritis Rheum 2002;47:432; with permission.)

IIFA using the HEp-2 cells or variants of this cell line (HEp-2000) is considered the gold-standard technique for the detection of ANA.[2] It involves serial dilutions of positive sera and visual determination of the staining pattern by fluorescent microscopy. The IIFA test is time consuming, is laborious, requires substantial expertise, and lacks specificity.[2] Indeed, depending on demographics, the population being studied, serum dilution, the cutoff used, and other variables of this assay, up to 25% of sera from apparently healthy individuals can be ANA positive.[2] One multicenter study reported that 31.7% of normal individuals were ANA positive at 1:40 dilution, which decreased to 13.3% at 1:80, 5.0% at 1:160 dilution, and 3.3% at 1:320 dilution.[4] Using HEp-2 substrate, titers of 1:80 or higher are usually considered positive.[5] ANA dilution at 1:160 has a sensitivity of 95% in patients with SLE and 87% sensitivity in patients with systemic sclerosis.[4]

Increased demand for ANA testing resulted in the development of novel diagnostic tests, including ELISA and other alternative methods like multiplex bead assays.[2] These tests are ones that are being ordered by primary care physicians (PCPs) more frequently, but one must keep in mind that the false negative and false positive ratios of these methods are different. In addition, several IIFA-positive patients (32.6%) have no detectable autoantibodies using the multiplex ANA screen, thereby suggesting several SLE patients may be ANA negative by this method.[6] Thus, if the clinical suspicion is strong, it is mandatory to perform IIFA.[2]

ANA testing plays an important role in the diagnoses of SLE. It is a part of clinical criteria and is a useful test (LR+ of 2.2) to obtain, if there is high clinical suspicion of SLE (**Table 4**). However, because of the low prevalence of SLE in the general population, most people with positive ANA do not have SLE. A negative ANA by IIFA is a very useful test, because it helps to rule out SLE because of its high sensitivity and negative LR of 0.1.[7] In systemic sclerosis, ANA is not a part of clinical criteria but is a useful test (LR− of 0.27) and should be performed in patients who are suspected of having systemic sclerosis, because a negative result should prompt the consideration of other fibrosing conditions (see **Table 4**).[7]

Other conditions in which ANA can be positive but does not have any proven value for diagnosis, monitoring, or prognosis are rheumatoid arthritis (RA), multiple sclerosis, thyroid disease, infectious diseases, fibromyalgia, and malignancies; thus, it is a not a useful test to order in patients with these conditions.[7]

Recently, the ability to order reflex or cascade testing, in which panels of tests for various other specific ANA-associated autoantibodies like double-stranded DNA (dsDNA), smith (Sm), ribonucleoprotein (nRNP), Ro (SS-A), La (SS-B), anticentromere, Scl-70, and Jo-1 are performed, has become available. Such tests are performed without regard to clinical characteristics, leading to increased costs and erroneous diagnoses.[5] If the ANA is negative by multiplex bead assay, by definition the specific autoantibodies will be negative because it is the same test.

ANA subtypes that can help establish and confirm the diagnosis of SLE include anti-dsDNA and anti-Sm.[5] A positive anti-Sm will likely have a large impact on the pretest

Table 4
Diagnostic utility of antinuclear antibody in autoimmune diseases

Condition	Sensitivity, %	Specificity, %	LR+	Utility	LR−	Utility
SLE	93–100	57	2.2	Useful	0.1	Very useful
SSc	85	54	1.86	Not useful	0.27	Useful
Sjögren syndrome	48	52	0.99	Not useful	1.01	Not useful
Inflammatory myositis	61	63	1.67	Not useful	0.61	Not useful
RA	41	56	0.93	Not useful	1.06	Not useful

Abbreviation: SSc, systemic sclerosis.

Modified from Solomon DH, Kavanaugh AJ, Schur PH. Evidence-based guidelines for the use of immunologic tests: antinuclear antibody testing. Arthritis Rheum 2002;47:437; with permission.

probability, substantially increasing the posttest probability of the diagnosis of SLE; thus, in the setting of some clinical suspicion of SLE, a positive anti-Sm strongly supports the diagnosis and is a very useful test to obtain, but a negative result cannot exclude the diagnoses (**Table 5**).[8]

Anti-dsDNA antibodies are useful to confirm the diagnosis for a patient whose clinical presentation already suggests a reasonable pretest likelihood of SLE being present (5% or more). It is a very useful test (LR+ of 16.3) and will substantially increase the posttest probability.[9] The presence of anti-dsDNA must always be interpreted in the context of the complete clinical picture, and also test characteristic should be taken into consideration because different assays have different characteristics as mentioned earlier. For example, the presence of anti-dsDNA and proteinuria or symptoms should prompt the clinician to investigate the presence of lupus nephritis or active lupus, but in an asymptomatic SLE patient, elevated anti-dsDNA may not indicate active SLE. Tests for antibodies to single-stranded DNA are of no clinical utility and should not be ordered.[5]

Antibodies to ribonucleoproteins, Ro (SS-A) and La (SS-B), are generally associated with SLE and Sjögren syndrome and are not routinely recommended without clinical suspicion. Anti-Ro (SS-A) is detected in approximately 35% to 60% of SLE patients and 40% to 90% in Sjögren syndrome patients and is associated with specific clinical features like photosensitivity, dry eyes and dry mouth, thrombocytopenia, and subacute

Table 5
Diagnostic utility of subserologies based on condition

ANA Subtype	Condition	Sensitivity, %	Specificity, %	LR+	Utility	LR−	Utility
Anti-Sm	SLE vs healthy controls	24	98	>10	Very useful	0.97	Not useful
	SLE vs other rheumatic diseases	30	96	26.5	Very useful	0.48	Useful
Anti-RNP	SLE vs other disease controls	27	82	1.2	Not useful	0.93	Not useful
	MCTD	71–100	84–100	7.14	Very useful	0	Very useful
Anti-dsDNA	SLE vs healthy controls and other diseases	57.3	97.4	16.3	Very useful	0.49	Useful
	Prognosis (overall activity of SLE)	66	66	4.14	Very useful	0.51	Not useful
	Lupus nephritis	65	41	1.7	Not useful	0.76	Not useful
Anti-centromere	SSc vs healthy controls	33	99.9	327	Very useful	0.7	Not useful
	SSc vs other CTDs	31	99.9	12.5	Very useful	0.7	Not useful
	Limited cutaneous (ACR classification)	44	93	6.1	Very useful	0.6	Not useful
Anti-Scl-70	SSc vs normals	20	100	25	Very useful	0.8	Not useful
	SSc vs other CTDs	26	99.5	>15	Very useful	1.5	Not useful
	Diffuse cutaneous involvement	37	82	2.0	Useful	0.8	Not useful
	Predicting pulmonary fibrosis	45	81	2.3	Useful	0.7	Not useful

Abbreviations: CTD, connective tissue disease; MCTD, mixed connective tissue disease.
Data from Refs.[8–10]

cutaneous lupus.[5] A very important correlation with anti-Ro is in neonatal lupus, because maternal immunoglobulin G (IgG) antibodies can cross the placenta. Symptoms in the neonate include rashes, and risk of congenital complete heart block. Hence, women with SLE considering pregnancy should be screened for anti-Ro (SS-A).[5]

Anticentromere and anti-Scl70 antibody testing is very useful in diagnosing systemic sclerosis and will have a large impact on posttest probability because they are rarely found in other connective tissue diseases or in healthy individuals (see **Table 5**).[10] Determination of anticentromere early in course of systemic sclerosis is very useful in predicting limited cutaneous involvement (distal to elbows) (LR+ of 6.1). Anti Scl-70 by immunodiffusion method is useful in predicting development of diffuse cutaneous involvement (LR+ of 2.0) and pulmonary fibrosis (LR+ of 2.3) (see **Table 5**).

Rheumatoid factor RF is any class of immunoglobulin, IgM, IgG, or IgA, that recognizes IgG.[11,12] The IgM-IgG RF subtype is the most widely used serologic marker of RA and is a part of the 2010 American College of Rheumatology/European League against Rheumatism (ACR/EULAR) criteria for the classification of RA.[13]

The prevalence of RF in RA is around 70% to 90%.[14] RF titers have been shown to correlate with both the severity of arthritis and progressive joint destruction.[14] Many studies have shown that the presence of elevated levels of RF precede the symptoms of RA.[12] RF-positive patients are also more likely to develop extra-articular manifestations, including rheumatoid nodules, Felty syndrome, and secondary Sjögren syndrome.[15] Different methods for rheumatoid factor detection include latex agglutination test, nephelometry, and ELISA.[12]

If a patient is presenting with chronic symmetric polyarticular joint swelling, RF is a very useful test (LR+ of 4.42) (**Table 6**) to obtain in order to support the diagnoses of RA. RF titers greater than 50 U/mL are more specific and will substantially increase the posttest probability of RA (LR+ of 20.5).[16] However, RF alone should not be used to make diagnoses of RA because it is present in many other conditions listed in **Table 7**.

Anticitrullinated Cyclic Peptide Antibody

Many novel antibodies are detected in association with RA, including antiperinuclear factor, antikeratin, and antifillagrin antibodies. All of these antibodies target citrullinated peptides. Citrullination is carried out by enzyme peptidylarginine deiminase.[12] Several citrullinated proteins are present in RA; hence, they have been identified as targets of highly RA-specific autoantibodies.[17] Anticyclic citrullinated peptide (anti-CCP) can be detected by ELISA, which is currently being ordered by PCPs.[17]

Table 6
Diagnostic utility of rheumatoid factor and anti-cyclic citrullinated peptide in rheumatoid arthritis

Autoantibodies	Sensitivity	Specificity	LR+	Utility	LR−	Utility
IgMRF (overall)	69%	85%	4.86	Very useful	0.38	Useful
Anti-CCP	67%	95%	12.46	Very useful	0.36	Useful
RF value	>20 U/mL		5.0	Very useful	0.53	Not useful
	>50 U/mL		11.3	Very useful	0.57	Not useful

Data from Nell VP, Machold KP, Stamm TA, et al. Autoantibody profiling as early diagnostic and prognostic tool for rheumatoid arthritis. Ann Rheum Dis 2005;64:1731–6; and Nishimura K, Sugiyama D, Kogata Y, et al. Meta-analysis: diagnostic accuracy of anti-cyclic citrullinated peptide antibody and rheumatoid factor for rheumatoid arthritis. Ann Intern Med 2007;146:797–808.

Table 7
Conditions with positive rheumatoid factor

Condition	Frequency of Occurrence (%)
Rheumatologic conditions	
RA	**60–80**
Juvenile chronic arthritis	15
Psoriatic arthritis	<15
SLE	30
Primary Sjögren syndrome	**70**
Mixed connective tissue disease	25
Polymyositis/dermatomyositis	20
Infections	
Subacute bacterial endocarditis,	**40**
Tuberculosis	15
Syphilis	8–37
Viral infections (hepatitis A, B, and C)	25
EBV/CMV	20
Coxsackie B	15
Dengue	10
HIV	10–20
Measles	8–15
Rubella	15
Parasitic infections (Chagas)	15–25
Malaria	15–18
Other conditions	
Cryoglobulinemia	**70**
Liver cirrhosis	25
Chronic interstitial lung disease	25

Modified from Dörner T, Egerer K, Fiest E, et al. Rheumatoid factor revisited. Curr Opin Rheumatol 2004;16:251; with permission.

Anti-CCP is a very useful test and is a more specific test to perform, if there is high pretest probability of RA (see **Table 6**).[18] However, anti-CCP can also be detected in many other conditions, including psoriatic arthritis, SLE, Sjögren syndrome, inflammatory myopathy, and active tuberculosis.[16] In patient with undifferentiated arthritis, a positive anti-CCP can indicate that the patient may develop RA and be at higher risk for rapid joint destruction.[17,19] Anti-CCP is also a part of the 2010 ACR/EULAR criteria for the classification of RA.[13]

The positive predictive value of anti-CCP is 95%, if a high-risk population is being tested rather than screening the entire population.[17] Negative test results for both anti-CCP and RF are useful when excluding the diagnosis of RA rather than testing of either antibody alone.[15]

Erythrocyte Sedimentation Rate and C-Reactive Protein

An acute-phase protein is defined as one in which plasma concentration increases or decreases by at least 25% during inflammatory disorders. The changes in concentrations of acute-phase proteins are due largely to changes in their production by hepatocytes through stimulation by cytokines. Conditions that lead to changes in the plasma concentration of acute-phase proteins include infection, trauma, surgery, burns, tissue infarction, rheumatologic diseases, crystal-induced inflammatory conditions, and advanced cancers. Various plasma proteins that increase in concentration during inflammation are complements, fibrinogen, plasminogen, alpha-1 chymotrypsin, ceruloplasmin, haptoglobin, C-reactive protein (CRP), serum amyloid A, fibronectin, fibrinogen, and ferritin. Currently, the most widely used indicators of the response of acute-phase proteins are erythrocyte sedimentation rate (ESR) and plasma CRP.[20]

ESR is a measure of quantity of red blood cells (RBCs) that precipitate in a tube in a defined time and is based on serum protein concentrations and RBC interactions with these proteins and is measured in millimeters per /hour. It depends on the concentration of various plasma proteins like fibrinogen, which increase during inflammation, promoting closer aggregation of RBC. Hence, the ESR is an indirect measure of inflammation.[21] Conditions that increase ESR are anemia, age, female sex, pregnancy, and hypercholesterolemia. ESR can be elevated with increased age, so it is important to adjust for age when interpreting normal range for ESR; in men, it is (age)/2 and in women it is (age + 10)/2.[21] ESR should not be used to make a diagnosis in patients with joint swelling of unclear cause; for example, to make a diagnosis of septic arthritis, ESR is not a useful test because it has LR+ of 0.84 and LR− of 2.4.[22]

Sensitivity of ESR in giant cell arteritis (GCA) is 84%; positive LR is 1.2, and negative LR is 0.35.[23] Hence, in a patient with clinical findings, a very high ESR is most likely due to GCA and justifies initiation of high-dose steroids because of high sensitivity. However, if the clinical evidence of GCA is low, a normal ESR reduces the probability of the disease because the negative LR is 0.35.[21]

CRP was named for its reactivity against C-polysaccharide in the cell wall of *Streptococcus pneumoniae*. Function of CRP is not clear, but it is thought that CRP is an innate immune protein that helps opsonize pathogens for phagocytosis and activates the complement system.[15] Serum CRP is a direct marker of inflammation, and its concentration changes more rapidly than ESR. It is unaffected by serum components and is a stable protein. Therefore, CRP may be a better reflection of current inflammation, but is not diagnostic of any disease.[15]

ANTINEUTROPHIL CYTOPLASMIC ANTIBODIES

Routinely, antineutrophil cytoplasmic antibodies (ANCAs) are detected by IIFA on ethanol-fixed neutrophils, and 2 general immunofluorescent patterns are observed: cytoplasmic (c-ANCA) and perinuclear (p-ANCA). The principal antigens for c-ANCA and p-ANCA are proteinase 3 (PR3) and myeloperoxidase (MPO), respectively, and are present in azurophilic granules of neutrophils. PR3/c-ANCA is strongly associated with granulomatous polyangiitis, with a sensitivity of 65% and specificity of 95% by IIFA method. ANCAs are detected in about 50% to 75% of patients with microscopic polyangiitis (MPA) with a predominant pattern of p-ANCA/MPO, and c-ANCA can also be detected in 25% of MPA patients. ANCA is found in 40% to 60% of patients with eosinophilic granulomatosis with polyangiitis, where both patterns can be found, although p-ANCA is more common. ANCA can also be seen in inflammatory bowel disease, primary biliary cirrhosis, SLE, RA, and juvenile inflammatory arthritis.[24]

Although a positive ANCA supports the diagnoses of vasculitis, physicians should keep in mind that vasculitis is very rare, and the probability that a patient with a positive ANCA test has the disease in question (posttest probability) depends on the patient's clinical manifestations. Tissue biopsy remains the gold standard to diagnose vasculitis given the high toxicity of treatment regimens used in vasculitis.[24]

SUMMARY

Clinicians should not rely on rheumatologic tests to make a diagnosis of a rheumatologic disease but rather take into consideration the prevalence and the posttest probability of a given disease in the right clinical settings.

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